

Digital Logic Design

Midterm

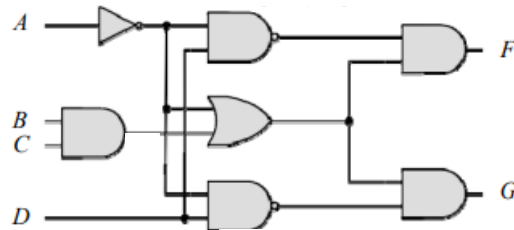
2025/04/28

姓名：

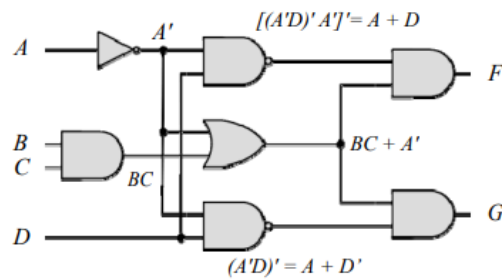
學號：

系級：

- Obtain the simplified Boolean expressions for output F and G in terms of the input variables in the following circuit. (10%)

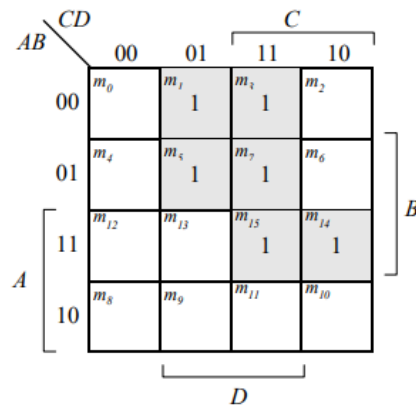


4.2

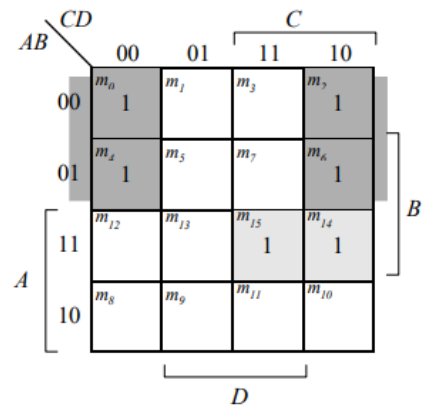


$$F = (A + D)(A' + BC) = A'D + ABC + BCD = A'D + ABC$$

$$F = (A + D')(A' + BC) = A'D' + ABC + BCD' = A'D' + ABC$$



$$F = A'D + ABC + BCD = A'D + ABC$$



$$G = A'D' + ABC + BCD' = A'D' + ABC$$

$$F=G$$

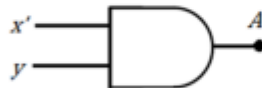
2. Design a combinational circuit with three inputs X , Y and Z , and three outputs A , B and C . When the binary input is 0, 1, 2, or 3, the binary output is twice the input. When the binary input is 4, 5, 6, or 7, the binary output is half the input. (20%)

4.5

x	y	z	A	B	C
0	0	0	0	0	0
0	0	1	0	1	0
0	1	0	1	0	0
0	1	1	1	1	0
1	0	0	0	1	0
1	0	1	0	1	0
1	1	0	0	1	1
1	1	1	0	1	1

		A			
		y			
yz	x	00	01	11	10
	0	m_0	m_1	m_3	m_2
	0	0	0	1	1
	1	m_4	m_5	m_7	m_6
	1	0	0	0	0

$$A = x'y$$



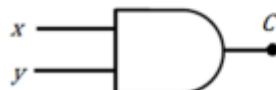
		B			
		y			
yz	x	00	01	11	10
	0	m_0	m_1	m_3	m_2
	0		1	1	
	1	m_4	m_5	m_7	m_6
	1	1	1	1	1

$$B = x + z$$



		C			
		y			
yz	x	00	01	11	10
	0	m_0	m_1	m_3	m_2
	0				
	1	m_4	m_5	m_7	m_6
	1			1	1

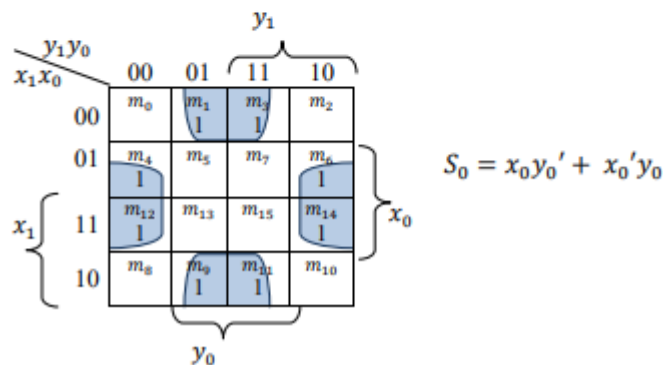
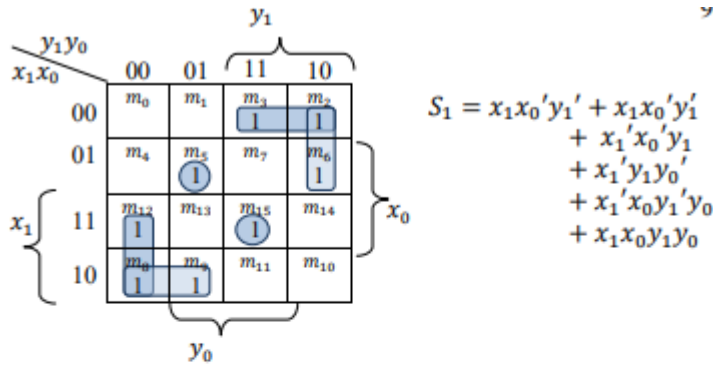
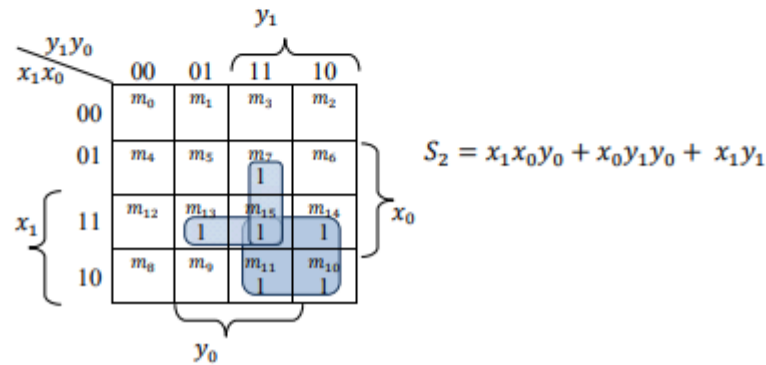
$$C = xy$$



3. Design a combinational circuit that adds two 2-bit numbers. The circuit outputs a three bit number that represents the arithmetic sum of the inputs. (15%)

4.21

x_1	x_0	y_1	y_0	S_2	S_1	S_0
0	0	0	0	0	0	0
0	0	0	1	0	0	1
0	0	1	0	0	1	0
0	0	1	1	0	1	1
0	1	0	0	0	0	1
0	1	0	1	0	1	0
0	1	1	0	0	1	1
0	1	1	1	1	0	0
1	0	0	0	0	1	0
1	0	0	1	0	1	1
1	0	1	0	1	0	0
1	0	1	1	1	0	1
1	1	0	0	0	1	1
1	1	0	1	1	0	0
1	1	1	0	1	0	1
1	1	1	1	1	1	0



4. Design a BCD-to-decimal decoder using the unused combinations of the BCD code as don't-care conditions. (10%)

4.24

Inputs: A, B, C, D

$$D_0 = A'B'C'D'$$

$$D_1 = A'B'C'D$$

$$D_2 = B'CD'$$

$$D_3 = B'CD$$

$$D_4 = BC'D'$$

Outputs: $D_0, D_1, \dots, D_9$

$$D_5 = BC'D$$

$$D_6 = BCD'$$

$$D_7 = BCD$$

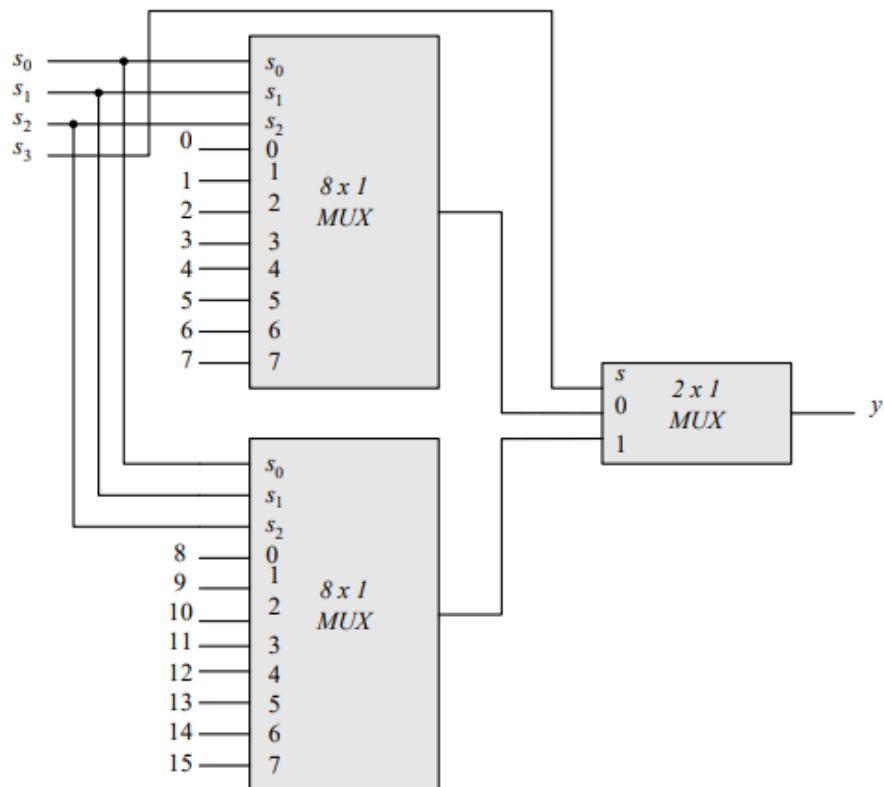
$$D_8 = AD'$$

$$D_9 = AD$$

		CD			
		00	01	11	10
AB	00	m_0 D_0	m_1 D_1	m_3 D_3	x D_2
	01	m_4 D_4	m_5 D_5	m_7 D_7	m_6 D_6
	11	m_{12} x	m_{13} x	m_{15} x	m_{14} x
	10	m_8 D_8	m_9 D_9	m_{11} x	m_{10} x

5. Construct a 16×1 multiplexer with two 8×1 and one 2×1 multiplexers. Use block diagrams. (10%)

4.31



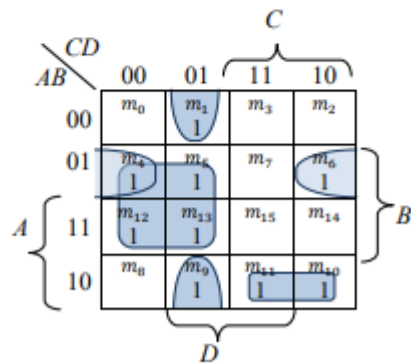
6. An 8×1 multiplexer has inputs A , B , and C connected to the selection inputs S_2 , S_1 , and S_0 , respectively. The data inputs I_0 through I_7 are as follows: (20%)

$$I_1 = I_7 = 0; I_2 = I_5 = I_6 = 1; I_0 = I_4 = D; \text{ and } I_3 = D'.$$

4.34 (a)

A	B	C	D	I	F
0	0	0	0	$I_0 = D$	0
0	0	0	1		1
0	0	1	0	$I_1 = 0$	0
0	0	1	1		0
0	1	0	0	$I_2 = 1$	1
0	1	0	1		1
0	1	1	0	$I_3 = D'$	1
0	1	1	1		0
1	0	0	0	$I_4 = D$	0
1	0	0	1		1
1	0	1	0	$I_5 = 1$	1
1	0	1	1		1
1	1	0	0	$I_6 = 1$	1
1	1	0	1		1
1	1	1	0	$I_7 = 0$	0
1	1	1	1		0

$$F = \Sigma(1,4,5,6,9,10,11,12,13)$$



$$F = A'BD' + AB'C + B'C'D + BC'$$

7. Using a decoder and external gates, design the combinational circuit defined by the following three Boolean functions: (15%)

$$F_1 = x'y' + xz$$

$$F_2 = xy'z' + yz$$

$$F_3 = x'y'z' + xy$$

4.28 (a)

$$F_1 = x'y' + xz = x'y'z' + x'y'z + xy'z + xyz = \Sigma(0,1,5,7)$$

$$F_2 = xy'z' + yz = xy'z' + x'yz + xyz = \Sigma(3,4,7)$$

$$F_3 = x'y'z' + xy = x'y'z' + xyz' + xyz = \Sigma(0,6,7)$$

